

Malhar Mahajan

(571) 581-4176 — malhar05@vt.edu — Ashburn, VA
linkedin.com/in/malhar-mahajan — github.com/malhar618 — malhar618.github.io

EDUCATION

Virginia Tech, Blacksburg, VA

Guidance, Control & Navigation Concentration

B.S. Aerospace Engineering May 2027

GPA 3.90/4.00

SKILLS

Languages: C/C++, Python, MATLAB

Interests: Simulink code generation, closed-loop simulation, multi-rate systems, estimation/filtering, controller validation

Tools: Git/GitHub, Linux, ROS 2, MAVSDK, ArduPilot/PX4, SITL/HITL, Siemens NX, Creo, SolidWorks

EXPERIENCE

Embedded Controls Co-op, Caterpillar Inc. (Technology Division) August 2025 – Present

- Refactoring OHT_07 IPS controls (ABS/TCS/DSC/MSE) into reusable Simulink blocks and AUTOSAR-ready interfaces across steering and propulsion variants.
- Built dual-channel rear-wheel traction-control plant co-simulation (Simulink + Dynasty) with Stateflow drive cycles that ramp vehicle speed and sweep road-friction coefficients to stress ABS/TCS/DSC slip response.
- Tuned brake/slip PID logic for legacy truck braking limits while owning V-model delivery for single-channel brake-based ABS: requirements, I/O, MIL, generated C, SIL/HIL, and machine-test planning.

Undergraduate Researcher, CARNATIONS Lab (Dr. Joerger) January 2026 – Present

- Developed UWB collaborative-navigation tooling at Virginia Tech for CARNATIONS: DWM1001 BLE/serial loggers, 3-link A1–B1/A2–B2/A3–B3 datasets, and GPS/UWB comparison plots.
- Implemented a 3-link relative-pose estimator (candidate roots + WLS/MAP prior) reinforced by GPS data; solved 81/81 GPS-aligned vehicle frames with 5.99 cm mean UWB fit RMSE.
- Calibrated SS-TWR with 7,553 outdoor samples across 1.65–30.39 m, reducing leave-one-distance-out RMSE from 8.9 cm to 3.7 cm.
- Extended COTS UWB range from 30 m to 100 m by rewriting firmware for lower bit rate and higher preamble length.

Simulations Intern, Caterpillar Inc. (GTMS)

May 2025 – August 2025

- Supported GTMS design decisions with thermal/CFD analyses and helped accelerate adoption of NX Nastran workflows across the simulation group.

Advanced Controls Systems Laboratory

August 2024 – August 2025

- Developed a 3-DOF thrust-stand testbed for small UAVs to support autonomous controller tuning; implemented a MATLAB genetic-algorithm workflow for parameter search and logged-data evaluation.
- Integrated Pixhawk 6C with an Odroid M1S running ROS 2 Galactic; built C++ nodes for Dynamixel-actuated moment cancellation using rigid-body inertia/moment models.
- Implemented IMU signal processing with high-pass filtering to derive acceleration from published position/velocity topics and validate estimates against logged data.

Team Lead, Avionics and Programming, GoAERO @VTech

August 2024 – September 2025

- Led avionics/software integration for a GoAERO UAV; defined ROS 2 interfaces between flight stack, sensors, and onboard compute.
- Built simulation-to-flight validation workflow (SITL/HITL + telemetry/command pipeline) to de-risk integration and support flight-test readiness across subteams.
- Contributed to NASA University Innovation Prize (\$30K) as part of a multidisciplinary aircraft design and autonomy team.

PROJECTS

Autonomous UAV

May 2024 – August 2024

- Built an autonomous quadrotor using a Jetson Nano and Intel RealSense D455; implemented real-time depth-based mapping/obstacle avoidance and target tracking in Python/OpenCV.
- Integrated MAVSDK flight-control interfaces and validated autonomy in JMAvSim, SITL, and HITL.